

SIEM LOG MONITORING & DETECTION LAB

Splunk Enterprise • Windows 11 • Sysmon • Universal Forwarder

Project type	Self-directed cybersecurity / SOC lab
Platform	Splunk Enterprise 10.4.2
Endpoint	Windows 11 virtual machine
Telemetry	Windows Security logs + Sysmon process telemetry
Detection use cases	Suspicious PowerShell execution; Windows brute-force activity
Status	Implemented, tested, and validated with controlled events
Evidence	Screenshots from the working lab environment

Prepared as a portfolio attachment for a Junior SOC / Cybersecurity Analyst CV.

1. Project Overview

This project demonstrates a practical SIEM monitoring workflow built from the ground up. Windows 11 telemetry was collected with Sysmon and the Splunk Universal Forwarder, ingested into Splunk Enterprise, investigated with SPL, and converted into working detections and alerts. The lab deliberately uses controlled test activity so that every detection can be traced from the originating endpoint event to the final SIEM alert.

Objectives

- Build an endpoint-to-SIEM telemetry pipeline using Windows 11, Sysmon, and the Splunk Universal Forwarder.
- Investigate Windows process-creation telemetry and identify suspicious PowerShell execution patterns.
- Analyze Windows Security Event ID 4625 for failed authentication activity.
- Create a threshold-based brute-force detection using a five-minute time window.
- Convert detections into Splunk real-time alerts and configure alert actions and throttling.
- Validate detections with controlled test activity and verify the resulting events in Splunk.

Environment

Windows 11 VM: endpoint generating telemetry

Sysmon: detailed process and endpoint telemetry

Splunk Universal Forwarder: log forwarding from the endpoint

Splunk Enterprise 10.4.2: indexing, searching, detection, and alerting

PowerShell: controlled activity generation and validation

VirtualBox: isolated virtualization environment

2. Telemetry Architecture

The implemented workflow follows the same basic pattern used in a SOC: generate telemetry at the endpoint, forward it to a central SIEM, normalize/search the data, identify suspicious behavior, and create an actionable alert.

Layer	Implementation	Purpose
Endpoint	Windows 11 VM	Generate real Windows security and process activity
Endpoint telemetry	Sysmon	Capture process creation and related endpoint events
Collection	Splunk Universal Forwarder	Forward telemetry to Splunk
SIEM	Splunk Enterprise 10.4.2	Index, search, correlate, and alert
Detection	SPL	Turn raw telemetry into security logic
Response signal	Real-time alerts	Surface suspicious behavior to an analyst

The screenshot displays the Splunk Enterprise interface on the left and a Windows 11 VM terminal on the right. The Splunk interface shows a search for Sysmon Event ID 1, resulting in 1,793 events. The search results table includes columns for _time, host, User, Image, and CommandLine. The Windows 11 VM terminal shows the output of the 'Get-Process' command, displaying a list of running processes with their handles, PIDs, PPIDs, WSIDs, CPU usage, IDs, SI values, and process names.

Handle	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
149	10	2224	11692	0.83	6568	0	AggregatorHost
398	22	9788	41632	0.67	6564	1	ApplicationFrameHost
133	10	1568	9196	2.36	940	0	conhost
443	22	8512	46184	0.95	6468	1	CrossDeviceResume
533	22	2816	6964	4.11	712	0	csrss
488	18	1944	7869	2.17	792	1	csrss
521	28	5456	27688	0.59	3988	1	ctfmon
1863	76	57480	91324	15.39	1180	1	dsm
151	10	2240	12508	0.09	4508	0	elevation_service
2561	82	53196	178544	12.72	5580	1	explorer
43	7	1500	4376	0.11	652	1	fontdrvhost
43	7	1480	4844	0.05	656	0	fontdrvhost
0	0	60	8	0	0	0	Idle
1335	25	7540	26296	8.39	948	0	lsass
0	0	480	87788	17.11	2320	0	Memory Compression
236	17	3184	6384	0.92	5820	0	MicrosoftEdgeUpdate
478	16	9184	25260	3.89	3392	0	HpDefenderCoreService
163	11	2820	13040	0.17	1224	1	msedge
175	11	9272	28284	0.20	5656	1	msedge
347	17	12332	28748	0.31	6852	1	msedge
1373	46	42832	137880	5.58	7156	1	msedge

Evidence: Splunk displaying Windows Sysmon Event ID 1 process-creation telemetry, including Image, CommandLine, ParentImage, and ParentCommandLine fields.

3. Detection Use Case #1 — Suspicious PowerShell Execution

The first detection focuses on suspicious PowerShell process creation. Instead of treating every PowerShell process as malicious, the search looks for command-line indicators commonly associated with abuse, including encoded commands, execution-policy bypass, and hidden-window execution.

Detection logic

```
index=windows sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1
| search Image="powershell.exe"
| search CommandLine="*-EncodedCommand*" OR CommandLine="*-enc *" OR CommandLine="*-ExecutionPolicy Bypass*" OR CommandLine="*-WindowStyle Hidden*" OR CommandLine="*-w hidden*"
| search User!="NT SERVICE\SplunkForwarder"
| eval detection="Suspicious PowerShell Execution"
| table _time host User detection Image CommandLine ParentImage ParentCommandLine
| sort - _time
```

Validation approach

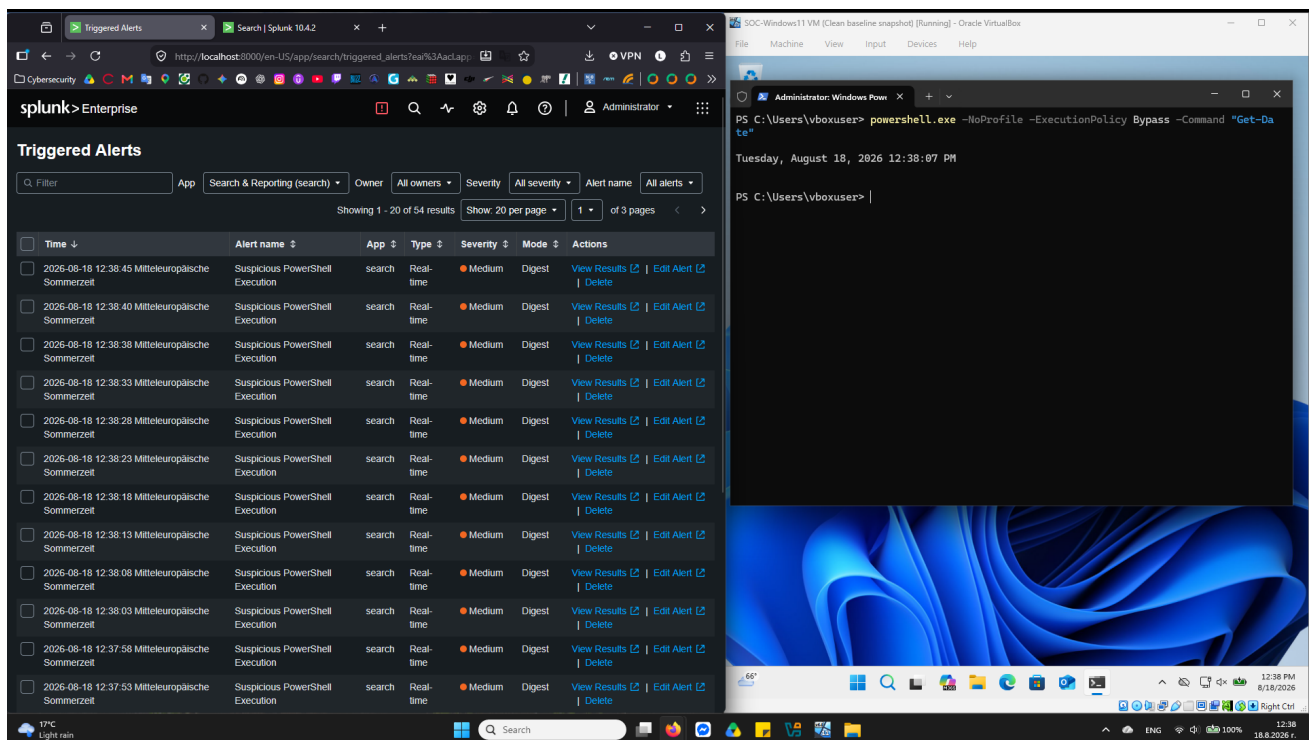
- Generated controlled PowerShell activity from the Windows VM.
- Used Sysmon Event ID 1 to confirm process creation was recorded.
- Inspected Image, CommandLine, ParentImage, and ParentCommandLine fields in Splunk.
- Used the detection search to distinguish suspicious PowerShell command-line patterns from normal PowerShell activity.
- Saved the search as a real-time Splunk alert named “Suspicious PowerShell Execution”.

The screenshot displays two windows side-by-side. The left window is the Splunk Enterprise interface, showing a search for suspicious PowerShell execution. The search query is: `index=windows sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1 | search Image="powershell.exe" | search CommandLine="*-EncodedCommand*" OR CommandLine="*-enc *" OR CommandLine="*-ExecutionPolicy Bypass*" OR CommandLine="*-WindowStyle Hidden*" OR CommandLine="*-w hidden*" | search User!="NT SERVICE\SplunkForwarder" | eval detection="Suspicious PowerShell Execution" | table _time host User detection Image CommandLine ParentImage ParentCommandLine | sort - _time`. The search results show a single event from 8/17/26 11:00:00.000 AM to 8/18/26 11:53:49.000 AM, with a detection of "Suspicious PowerShell Execution". The right window is a Windows VM console showing PowerShell commands and a process list. The commands executed are `powershell.exe -NoProfile -Command "Get-Date"` and `powershell.exe -NoProfile -Command "Get-Process"`. The process list shows various system processes, including `AggregatHost`, `ApplicationFrameHost`, `conhost`, `CrossDeviceResume`, `csrss`, `ctfmon`, `dwm`, `elevation_service`, `explorer`, `fontdrvhost`, `fontdrvhost`, `Idle`, `lsass`, `Memory Compression`, `MicrosoftEdgeUpdate`, `MpDefenderCoreService`, `msedge`, and `msedge`.

Evidence: A suspicious PowerShell event is visible in Splunk while the Windows VM simultaneously shows controlled PowerShell activity.

4. PowerShell Alert Validation

The alert was tested end-to-end rather than simply saved in Splunk. The screenshots show the alert being triggered after controlled PowerShell activity, providing evidence that the detection logic was connected to Splunk's alerting engine.



Evidence: Splunk Triggered Alerts view showing repeated “Suspicious PowerShell Execution” alerts generated by the test activity.

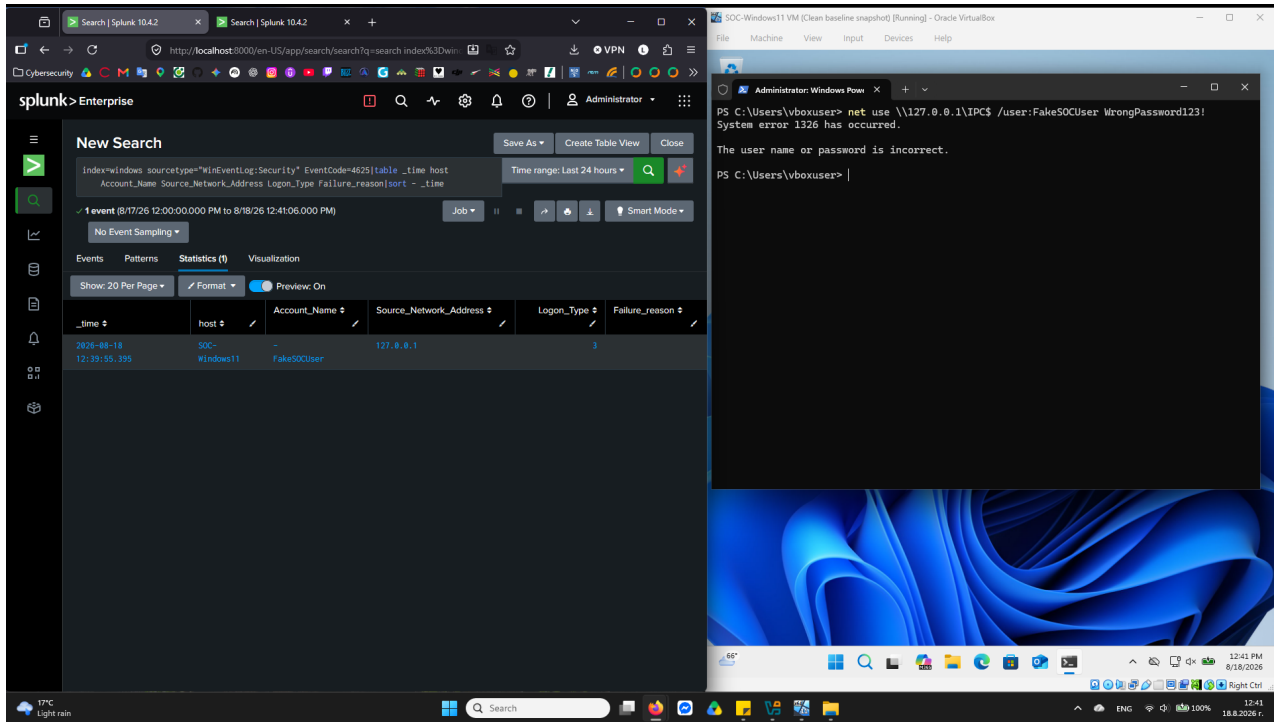
SOC relevance

This use case demonstrates how an analyst can move from raw process telemetry to a behavior-based detection. It also demonstrates why fields such as Image, CommandLine, ParentImage, and ParentCommandLine are valuable when investigating process execution.

5. Detection Use Case #2 — Windows Brute Force

The second use case detects repeated failed Windows logons. Windows Security Event ID 4625 was selected because it records failed authentication attempts. The lab generated controlled failed network logons and then examined the resulting telemetry in Splunk.

Initial event validation



Evidence: A Windows Security Event ID 4625 is visible in Splunk after a controlled failed network logon. The event includes the failed account, source address 127.0.0.1, and Logon Type 3.

Why Event ID 4625 matters

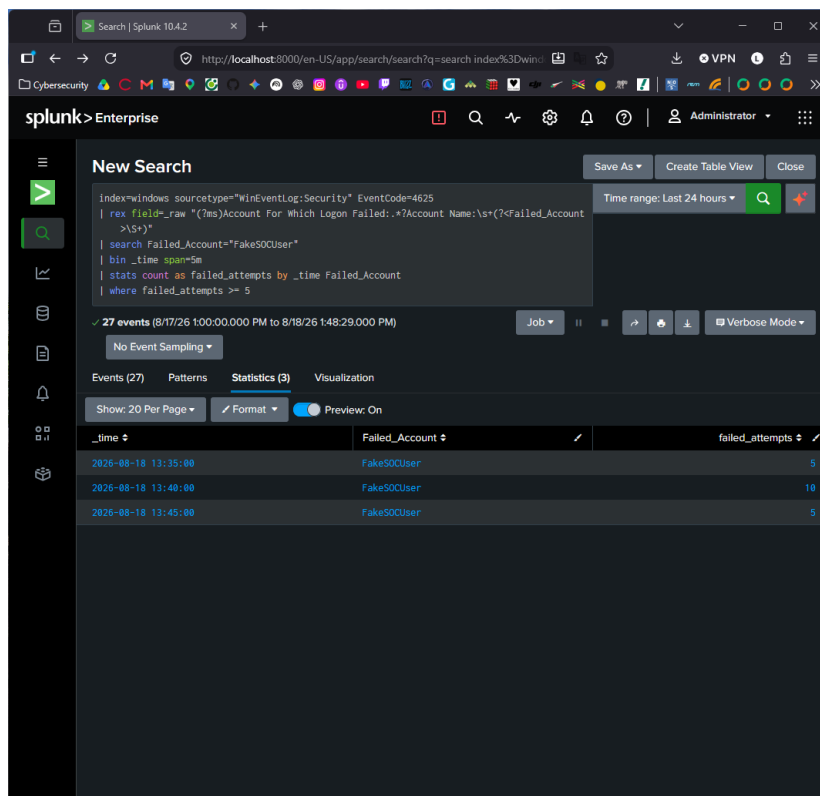
A single failed login is normally not sufficient to identify a brute-force attack. The detection therefore moves from single-event analysis to threshold-based correlation: multiple failures for the same account inside a defined time window are treated as suspicious.

6. Brute-Force Correlation Logic

The working SPL uses a five-minute time bucket and counts failed logons per extracted account:

```
index=windows sourcetype="WinEventLog:Security" EventCode=4625
| rex field=_raw "(?ms)Account For Which Logon Failed:.*?Account Name:\s+(?<Failed_Account>\S+)"
| bin _time span=5m
| stats count as failed_attempts by _time Failed_Account
| where failed_attempts >= 5
```

A custom rex extraction was used because the account name was not consistently exposed through the automatically extracted field set. This demonstrates a practical SIEM skill: extracting useful indicators directly from raw event data when field extraction is incomplete.



Evidence: Splunk groups failed authentication events into five-minute windows and shows multiple threshold breaches for FakeSOCUser.

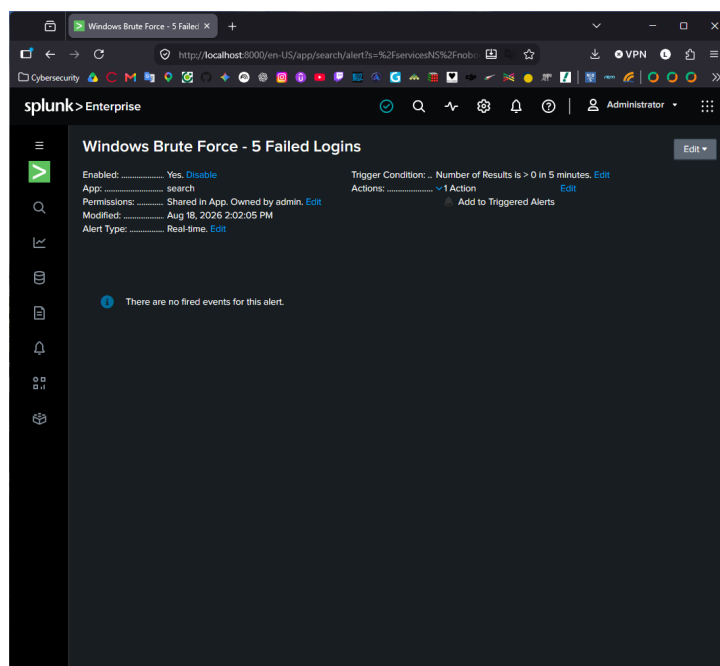
Controlled validation

- The test intentionally generated incorrect credentials so Windows would record Event ID 4625.
- The search initially confirmed five failed attempts in a five-minute window.
- The detection was then tested with additional attempts, including a separate test account, to verify the logic was not dependent on a single hard-coded username.
- The resulting count was visible directly in Splunk's Statistics view.

7. Alert Engineering and Tuning

The brute-force search was converted into a real-time Splunk alert named “Windows Brute Force - 5 Failed Logins”. The alert was configured to trigger when the search returned results and to add the event to Triggered Alerts.

Setting	Configured value
Alert name	Windows Brute Force - 5 Failed Logins
Alert type	Real-time
Trigger condition	Number of results > 0
Detection window	5 minutes
Threshold	At least 5 failed attempts
Trigger frequency	Once
Action	Add to Triggered Alerts
Severity	Medium
Throttle	Enabled — 5 minutes



Alert tuning lesson

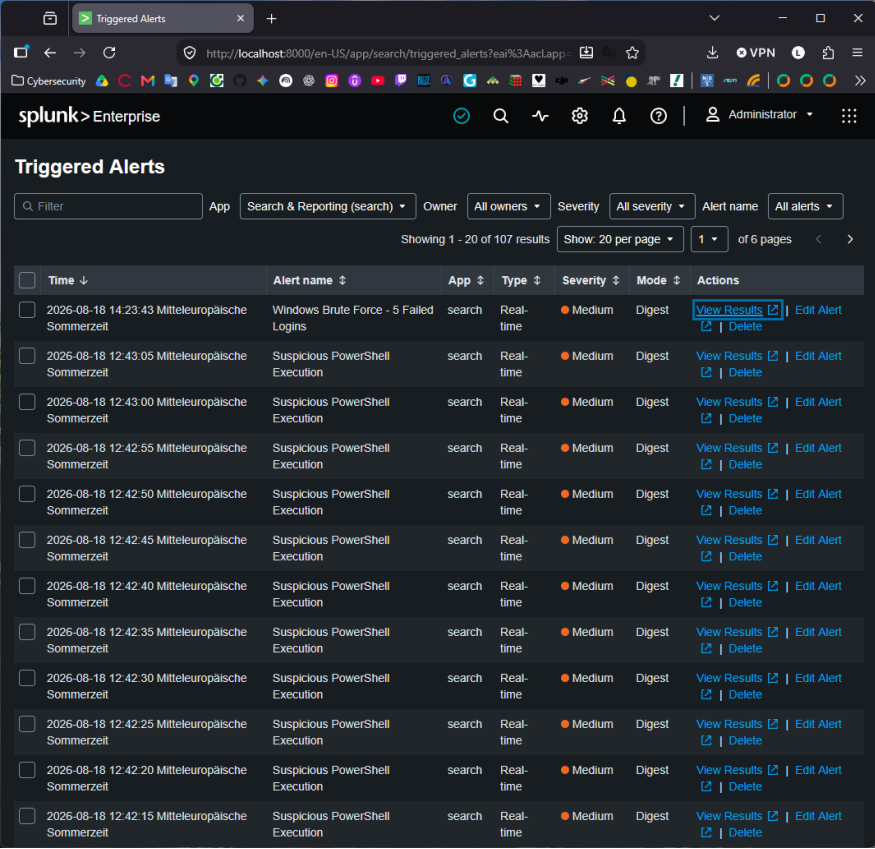
During testing, the real-time condition initially produced repeated alert activity. A five-minute throttle was enabled to suppress duplicate alerts after a trigger. This is representative of real SOC alert engineering: a technically correct detection can still require tuning to prevent alert noise and duplicate notifications.

8. Validation and Detection Hardening

A key improvement was removing the original hard-coded test-account filter from the brute-force detection. The final logic detects repeated failed authentication activity regardless of the username. This makes the detection more reusable and closer to a production-oriented rule.

Testing sequence

- Generate a known failed authentication event.
- Confirm Event ID 4625 in Splunk.
- Extract the failed account from the raw event.
- Aggregate failures over five minutes.
- Confirm the threshold is reached.
- Verify the Splunk alert fires.
- Enable throttling to reduce duplicate alert noise.
- Retest after removing the hard-coded username dependency.



The screenshot displays the 'Triggered Alerts' page in the Splunk Enterprise web interface. The interface includes a search bar, filters for App, Search & Reporting, Owner, Severity, and Alert name, and a table of triggered alerts. The table shows 107 results, with the first 10 displayed. The alerts are categorized by Time, Alert name, App, Type, Severity, and Mode. The first alert is 'Windows Brute Force - 5 Failed Logins' and the subsequent 9 are 'Suspicious PowerShell Execution'.

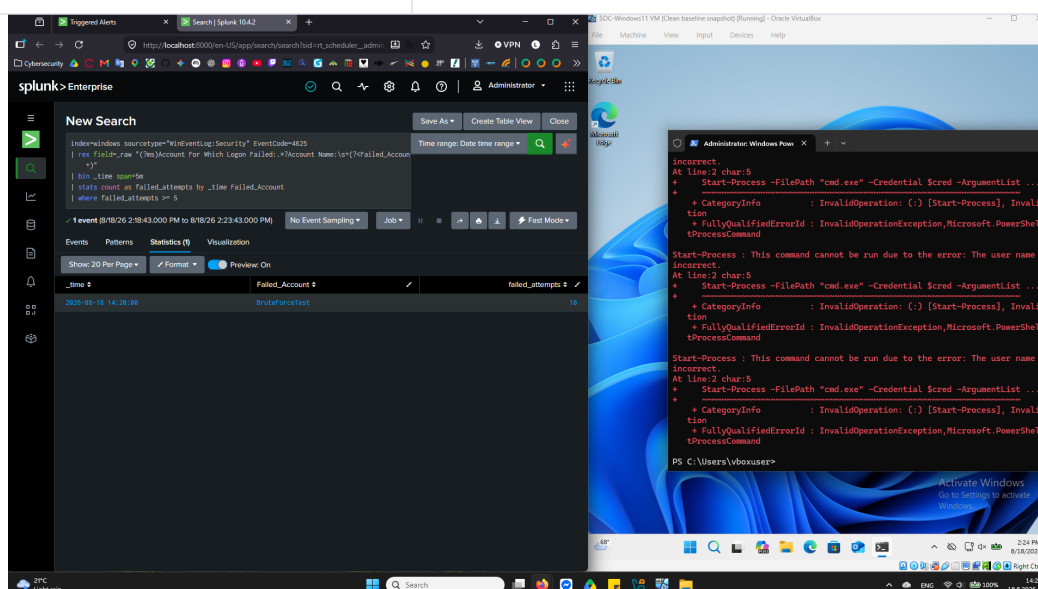
Time	Alert name	App	Type	Severity	Mode	Actions
2026-08-18 14:23:43 Mitteleuropäische Sommerzeit	Windows Brute Force - 5 Failed Logins	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:43:05 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:43:00 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:55 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:50 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:45 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:40 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:35 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:30 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:25 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:20 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete
2026-08-18 12:42:15 Mitteleuropäische Sommerzeit	Suspicious PowerShell Execution	search	Real-time	Medium	Digest	View Results Edit Alert Delete

Evidence: Splunk Triggered Alerts demonstrates that both the PowerShell and Windows brute-force detections are producing alert records.

9. Troubleshooting and Investigation Skills Demonstrated

The lab was not built by assuming that every command would immediately work. Several iterations were used to validate the telemetry and correct the detection logic.

Issue / observation	Resolution / lesson
Expected username fields were not consistently available	Used rex against _raw to extract Account Name.
Single failed login did not represent brute-force behavior	Changed the detection from single-event matching to threshold correlation.
Real-time alerts repeatedly fired	Enabled five-minute throttling to reduce duplicates.
Initial testing relied on a known test account	Removed the username-specific filter and retested the generic detection.
A credential-based PowerShell test produced a Start-Process error	Validated the failure as a test-command/credential issue rather than treating it as a SIEM failure.



Evidence: A controlled PowerShell credential test returned a Windows credential error. The event was investigated rather than incorrectly interpreted as a Splunk detection failure.

10. Results and Skills Demonstrated

The completed work demonstrates the following practical SOC/SIEM capabilities:

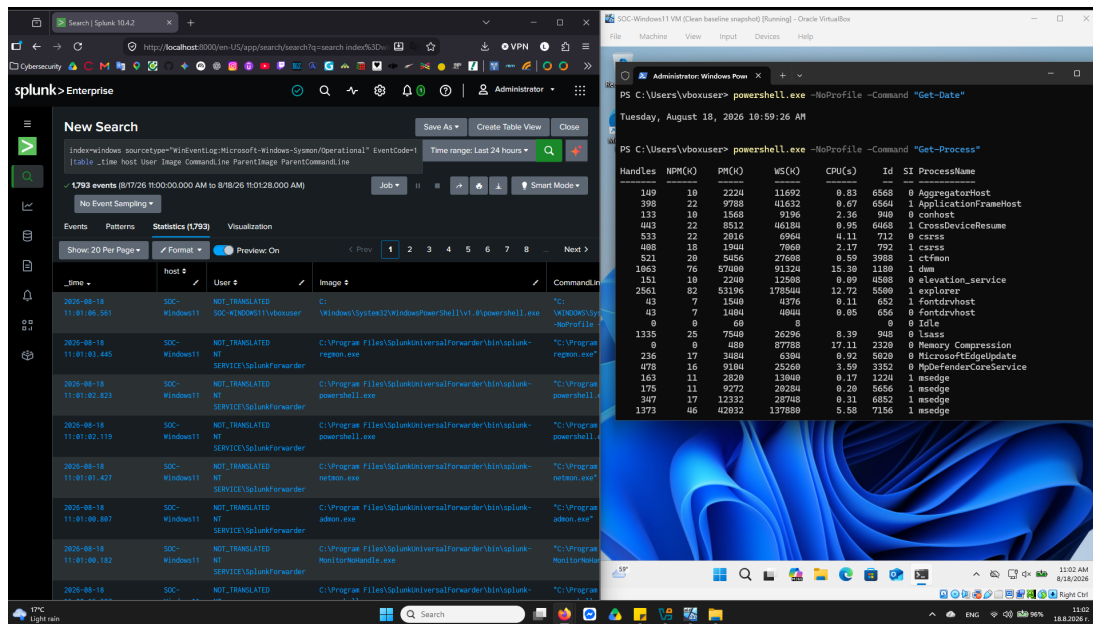
Area	Evidence of practical work
SIEM	Splunk Enterprise 10.4.2; SPL search, statistics, alerts, Triggered Alerts
Endpoint telemetry	Sysmon Event ID 1 process creation and Windows Security Event ID 4625
Log collection	Splunk Universal Forwarder from Windows endpoint
Detection engineering	Behavior-based PowerShell detection and threshold-based brute-force detection
Correlation	Five-minute aggregation by failed account
Field extraction	rex extraction from raw Windows event data
Alert engineering	Real-time alerts, severity, actions, and throttling
Validation	Controlled event generation and end-to-end verification
Troubleshooting	Investigating field extraction, test failures, alert noise, and detection tuning

Project outcome

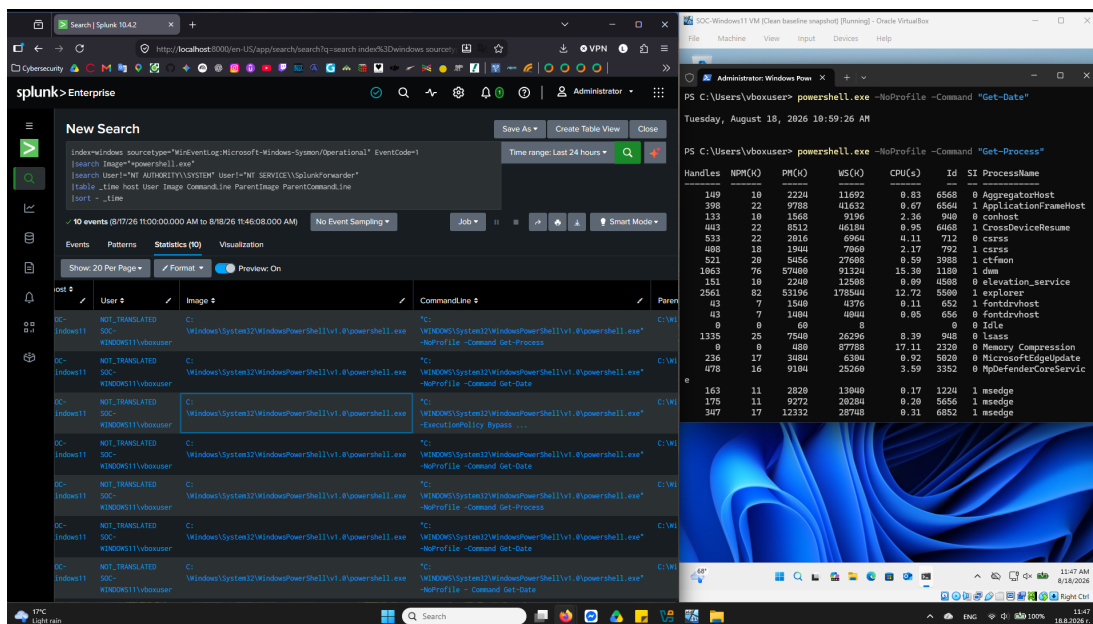
The Windows branch of the lab is operational: endpoint telemetry reaches Splunk, the data can be investigated with SPL, detections identify suspicious behavior, and real-time alerts can be generated and tuned. The next planned stage is to extend the same architecture to Linux/SSH telemetry and then build cross-platform correlation and a SOC dashboard.

Appendix — Selected Evidence

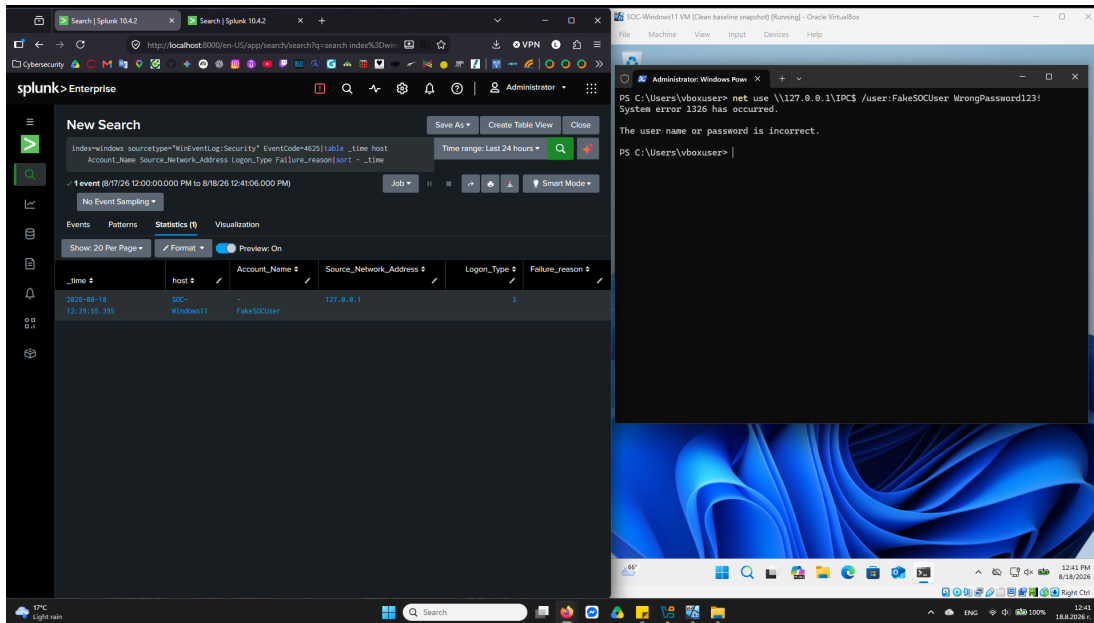
The following screenshots were retained as visual proof of the working environment, searches, events, and alerts.



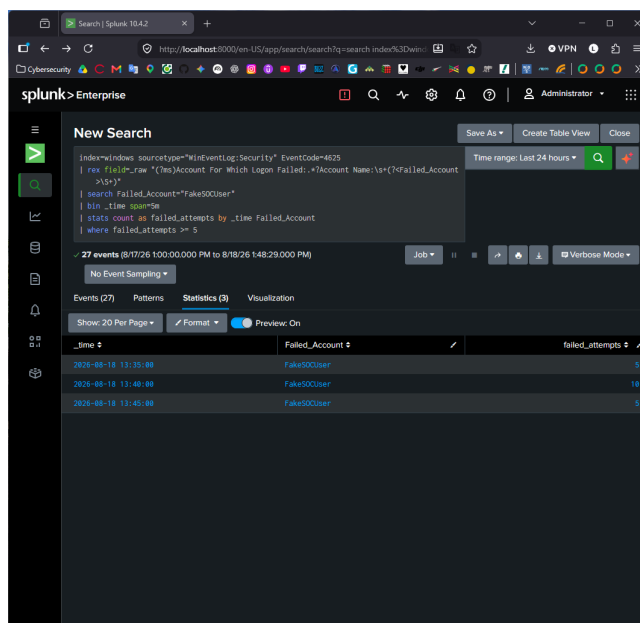
Evidence: Sysmon process-creation telemetry visible in Splunk.



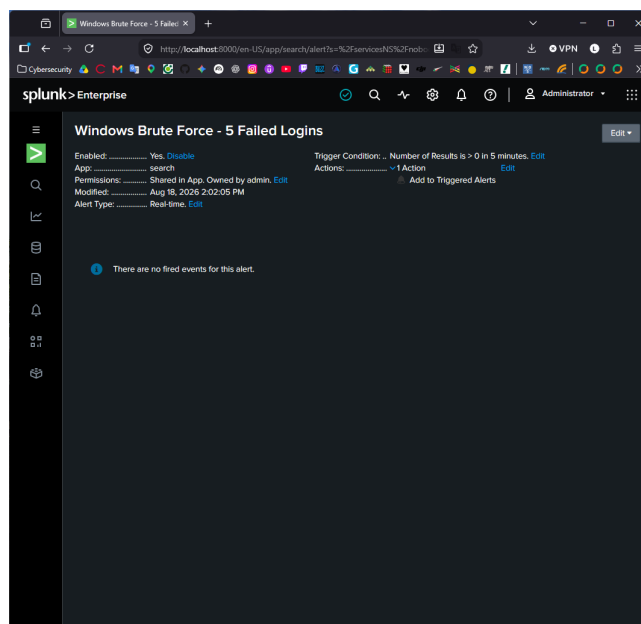
Evidence: PowerShell-related process events displayed with command-line and parent-process context.



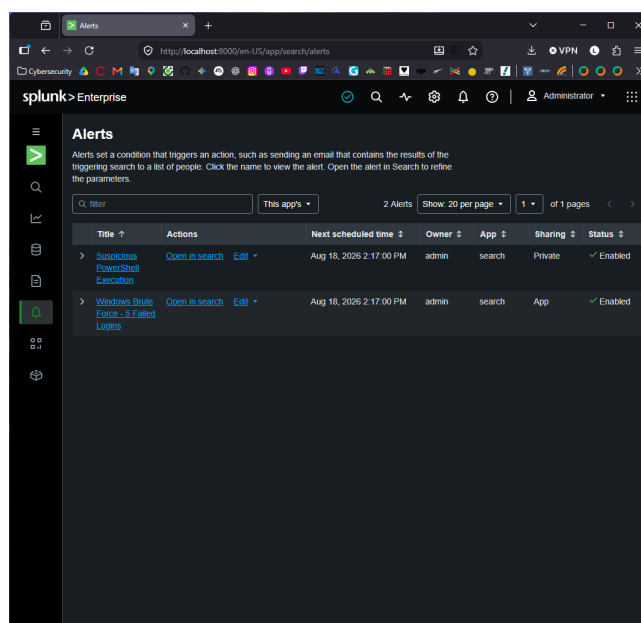
Evidence: Windows 4625 failed-logon event generated by controlled testing.



Evidence: Five-minute aggregation showing threshold breaches.



Evidence: Brute-force alert configuration and Triggered Alerts action.



Evidence: Splunk Alerts page showing enabled detections.